



What the Raised Hands Don't Say

A Survey of Comprehension Checks Against the Device-Free Hour Hand-Raise Tally

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The problem

Slop University's Device-Free Hour pilot renews its continuation case each term on one figure: how many hands go up. Whether a rising tally tracks what students can actually show they understood, rather than how often they reach for the ceiling, has not been checked.

Research questions

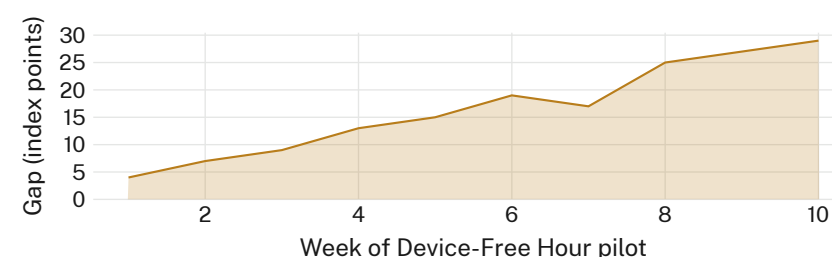
- Do session hand-raise counts predict comprehension-check pass rates?
- Do lecturing staff and students read the same session the same way?
- Does the tally-comprehension gap hold steady or widen across the pilot?

Study design

- n = 214 first-year students · 16 sessions · 4 courses · one teaching period
- Paired per session: TA-logged hand-raise tally · blind-scored 5-item comprehension check
- Parallel staff engagement-perception survey (n = 7 lecturing staff); scoring thresholds pre-registered

What we found

Session hand-raise counts explain 12% of the variance in comprehension-check pass rates ($r = 0.34$, $n = 214$). Staff engagement ratings track the tally ($r = 0.68$) more closely than either measure tracks comprehension.



The gap between the session hand-raise index and the comprehension-check pass rate widened from 4 points in week 1 to 29 points by week 10, tracking the tally's own rise almost exactly.

Implications & where next

The continuation case reads the tally as a stand-in for understanding; the comprehension checks say the two parted ways by week 4 and kept drifting. Staff ratings move with the raised hands, not the quiz. Next: pair the tally with a comprehension check by default, not as a one-term add-on.

Selected references

1. Dallimore, E. J., Hertenstein, J. H., & Platt, M. B. (2004). Classroom participation and discussion effectiveness: student-generated strategies. *Communication Education*, 53(1), 103–115. doi:10.1080/0363452032000135805
2. Beland, L.-P., & Murphy, R. (2016). Ill communication: technology, distraction & student performance. *Labour Economics*, 41, 61–76. doi:10.1016/j.labeco.2016.04.004
3. Black, P., & Wiliam, D. (1998). Assessment and classroom learning. *Assessment in Education: Principles, Policy & Practice*, 5(1), 7–74. doi:10.1080/0969595980050102
4. Caldwell, J. E. (2007). Clickers in the large classroom: current research and best-practice tips. *CBE — Life Sciences Education*, 6(1), 9–20. doi:10.1187/cbe.06-12-0205
5. Sabel, R., Nwosu, J., & Solberg, T. (2026). Pouched, not prevented: a quasi-experimental evaluation of numbered device-storage pouches and the Invigilation Readiness Index. Slop University. doi:10.5555/slop.zqrgbc

Comprehension checks scored blind to each session's hand-raise count; results reported in aggregate only.

We thank the lecturing staff who let a tally-keeper sit in on sessions already running device-free.

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